Data Structures and Algorithms

Data structures organize data for efficient access.
Time Complexity: O(1) < O(log n) < O(n) < O(n log n) < $O(n^2)$
Linear Structures: Arrays, Linked Lists, Stacks, Queues
Arrays: O(1) access, O(n) insertion/deletion
Linked Lists: O(n) access, O(1) insertion/deletion
Stacks: LIFO (Last In First Out)
Queues: FIFO (First In First Out)
Trees: Binary Search Tree, AVL Tree, Red-Black Tree
BST: Left < Root < Right, O(log n) average operations
Heaps: Complete binary tree with heap property
Graphs: Vertices and Edges, Adjacency Matrix/List
Sorting Algorithms:
- Merge Sort: O(n log n), stable
- Quick Sort: O(n log n) average, fastest in practice
- Heap Sort: O(n log n), in-place
Search Algorithms:
- Binary Search: O(log n), requires sorted data
- Hash Table: O(1) average case
Dynamic Programming: Overlapping subproblems + Optimal substructure
References: Cormen et al., Introduction to Algorithms